12) $f(x) = \dfrac{2-\sqrt{x}}{2+\sqrt{x}} = \dfrac{2-x^{\frac{1}{2}}}{2+x^{\frac{1}{2}}} = (2-x^{\frac{1}{2}})(2+x^{\frac{1}{2}})^{-1}$

$$f'(x) = \frac{-\frac{1}{2}x^{-\frac{1}{2}}(2+x^{\frac{1}{2}}) - \frac{1}{2}x^{-\frac{1}{2}}(2-x^{\frac{1}{2}})}{(2+x^{\frac{1}{2}})^2} =$$

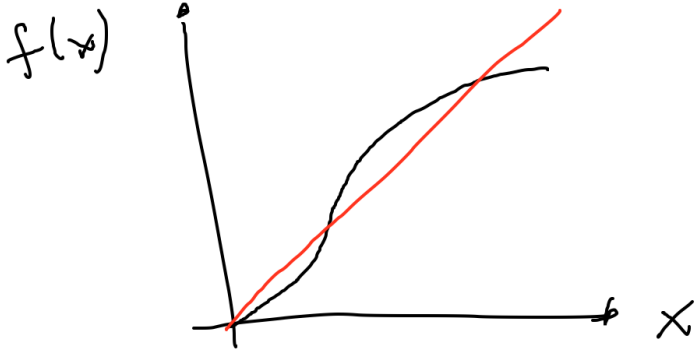
$$f'(x) = -\frac{1}{\cancel{2}}\,x^{-\frac{1}{2}}\,\frac{\left(\cancel{2+x^{\frac{1}{2}}}+\cancel{2-x^{\frac{1}{2}}}\right)}{(2+x^{\frac{1}{2}})^2} = \frac{-2\,x^{-\frac{1}{2}}}{(2+x^{\frac{1}{2}})^2}$$

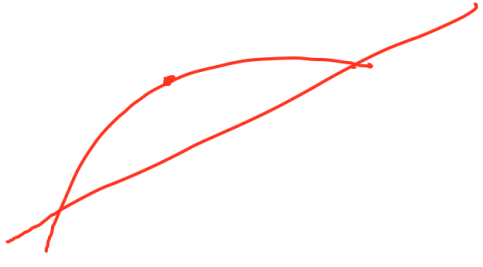
$$f'(x) = \frac{-2x^{-\frac{1}{2}}}{\left(2+x^{\frac{1}{2}}\right)^2}$$

$$f''(x) = \frac{\cancel{-2}\cdot\cancel{-\frac{1}{2}}\,x^{-3/2}\left(2+x^{\frac{1}{2}}\right)^{\cancel{2}\,1} - \cancel{2\left(2+x^{\frac{1}{2}}\right)^{1}}\cdot\cancel{\frac{1}{2}}\,x^{-\frac{1}{2}}\left(-\cancel{2}x^{-\frac{1}{2}}\right)}{\left(2+x^{\frac{1}{2}}\right)^{\cancel{4}\,3}}$$

$$f''(x) = \frac{x^{-\frac{3}{2}}\left(2+\sqrt{x}\right)+2x^{-1}}{\left(2+x^{\frac{1}{2}}\right)^3}$$

6) CONCAVIDADE / P.I

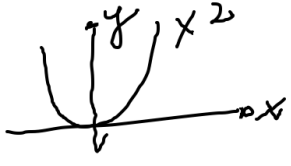


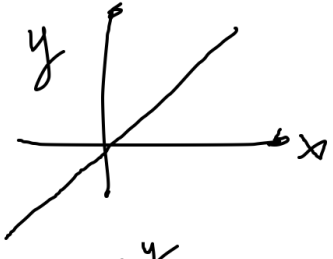


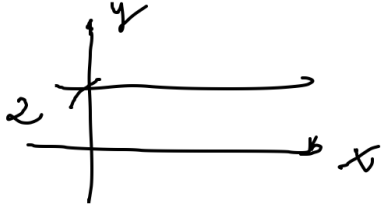
$f(x) = x^2$

$f'(x) = 2x$

$f''(x) = 2 > 0$





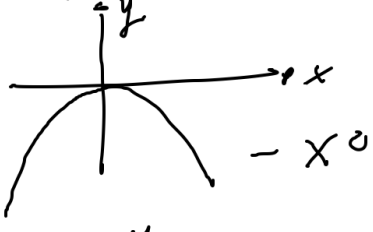


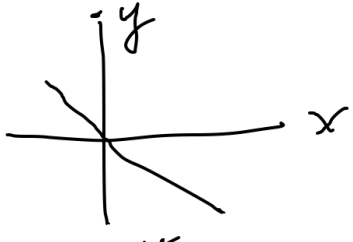
$f''(x) > 0$

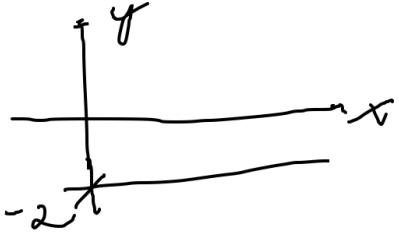
$f(x) = -x^2$

$f'(x) = -2x$

$f''(x) = -2$







$f''(x) < 0$

PONTO DE INFLEXÃO - PI

↓

onde ocorre a mudança de concavidade

Ex: $f(x) = x^3$

$f(x) = x^3 - 3x$

$f(x) = \sqrt[3]{x}$

$$f(x) = \begin{cases} 4 - x^2 & \text{pe } x \leq 1 \\ 2 + x^2 & \text{pe } x > 1 \end{cases}$$

$f(x) = x^3 - 3x^2 + 3$

$$\begin{cases} 4.5 - 1-4 \\ 4.6 - 1-4 \end{cases}$$